The Geometric Signature of Primes in Farey Sequences

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Abstract

We study how the uniformity of the Farey sequence changes each time a new integer is included—a per-step perspective that appears not to have been previously investigated. Primes and composites play opposite roles: composites improve uniformity while primes disrupt it, with the direction of each prime's effect controlled by the Mertens function.

We prove exact algebraic identities for the per-step discrepancy change, including a four-term decomposition, an injection principle showing each new fraction enters a distinct gap, and a dominance result for the fraction $1/p$. These identities reveal that the sign of the prime disruption correlates strongly ($r = 0.77$) with the leading zero of the Riemann zeta function, connecting Farey geometry to the Chebyshev bias framework of Rubinstein and Sarnak.

A spectral function built from the per-step data detects the first three nontrivial zeta zeros from Farey discrepancy data alone. All core identities are formally verified in Lean 4 (422 results, zero `sorry`).

Keywords: Farey sequence, Mertens function, Chebyshev bias, zeta zeros, rank deviation, Ramanujan sums, formal verification, Lean 4.
MSC 2020: 11B57, 11M06, 11N37, 11A25, 11Y35.

1 Introduction

1.1 Farey sequences and their discrepancy

For a positive integer N, the *Farey sequence* $\mathcal{F}_N$ is the ascending list of fractions a/b with $0 \le a \le b \le N$ and $\gcd(a, b) = 1$. Its cardinality is

$$|\mathcal{F}_N| = 1 + \sum_{k=1}^{N} \varphi(k). \tag{1}$$

Writing $n = |\mathcal{F}_N|$ and listing elements as $f_0 < f_1 < \cdots < f_{n-1}$, the *squared rank deviation* (which we call the *wobble*) is

$$W(N) \;=\; \sum_{j=0}^{n-1} \left(f_j - \frac{j}{n} \right)^2. \tag{2}$$

Remark 1.1 (Interval vs. circle)**.** The Farey sequence lives on the *closed interval* $[0, 1]$, with $f = 0$ and $f = 1$ as distinct elements. Throughout this paper we use circle diagrams (identifying $[0, 1)$ with $\mathbb{R}/\mathbb{Z}$) as a visualization aid, but the discrepancy the wobble definition and all identities in

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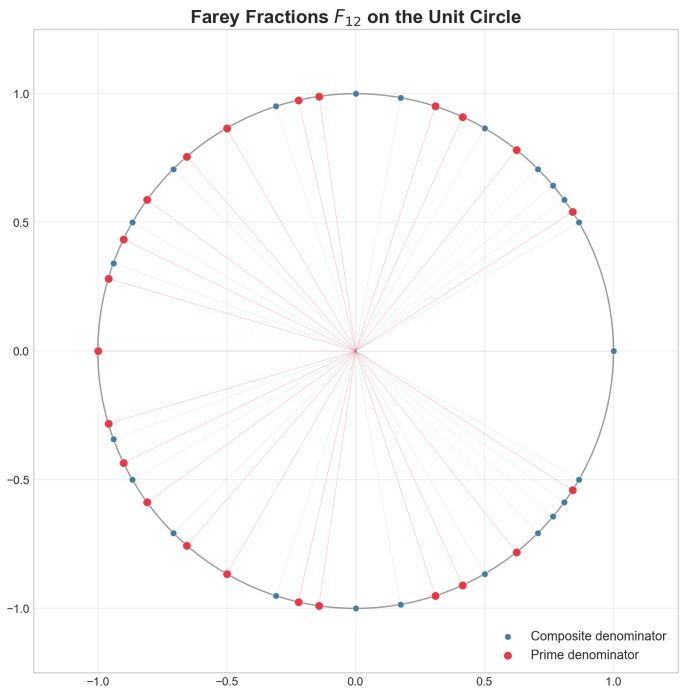

Figure 1: The Farey sequence $\mathcal{F}_{12}$ on the unit circle: each fraction a/b is placed at angle $2\pi a/b$. The pattern is strikingly uniform, with the largest gaps near 0 and 1 (top of the circle) where the denominators are smallest.

Section 3 are stated and proved on the interval, where the two boundary points $f = 0$ and $f = 1$ contribute separately. In particular, the $+2$ in the bridge identity (Theorem 3.1) arises from these two distinct boundary contributions.

1.2 The Franel–Landau theorem

Franel [6] and Landau [7] proved that the Riemann Hypothesis is equivalent to the bound $\sum_{j=0}^{n-1} |f_j - j/n| = O(N^{1/2+\varepsilon})$ for every $\varepsilon > 0$, where $n = |\mathcal{F}_N|$. Equivalently (see Hardy–Wright [11], Ch. XVIII), RH holds if and only if $W(N) = O(N^{-1+\varepsilon})$. Thus the quantity $W(N)$ defined above—which we call the *wobble* (our terminology for the squared rank deviation)—encodes information about the non-trivial zeros of the Riemann zeta function. Recent work by Karvonen and Zhigljavsky [15] studies the maximum mean discrepancy of Farey sequences, providing additional quantitative connections between Farey uniformity and analytic number theory.

1.3 Motivating observation: primes fill voids uniformly

Place the Farey fractions of $\mathcal{F}_N$ as points on the unit circle at angles $2\pi f$. As N grows, new fractions appear and the circle fills in. The first author observed a striking asymmetry: when a *prime* p is added, every one of the $p-1$ new fractions k/p lands on the circle at perfectly equispaced positions, filling voids uniformly across the entire circle. This happens because every integer below a prime is coprime to it ($\varphi(p) = p - 1$), so no position is skipped. Composites, by contrast, skip positions whose numerators share a factor with the denominator, leaving voids in some regions while clustering in others.

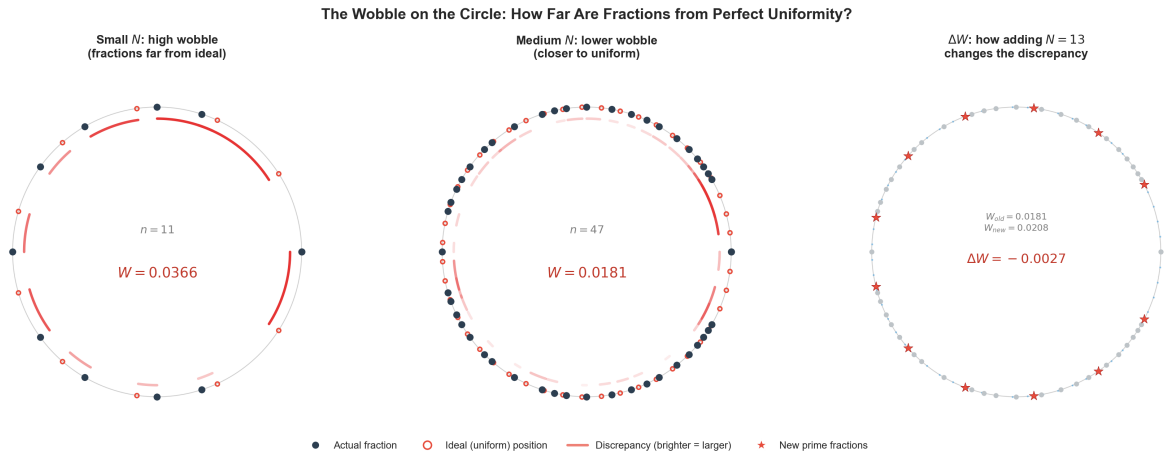

Figure 2: The wobble visualized on the circle. Dark dots show actual Farey fractions; red circles show ideal (equally spaced) positions. Red arcs connect each fraction to its ideal position—brighter arcs mean larger discrepancy. **Left:** $\mathcal{F}_5$ has high wobble (large gaps between actual and ideal). **Center:** $\mathcal{F}_{12}$ has lower wobble (fractions closer to uniform). **Right:** adding prime $p = 13$ (red stars) changes the ideal positions and the wobble.

Moreover, by the standard mediant property of Farey sequences, each new fraction k/p is the *mediant* of its two neighbors in $\mathcal{F}_{p-1}$—the geometrically optimal insertion point. This means primes do not merely add points; they add them in the most balanced way possible.

These geometric observations led to a natural question: does the uniformity of the Farey sequence on the circle improve or worsen at each step, and what controls the answer?

1.4 Per-step decomposition: a new perspective

Despite decades of work on the Franel–Landau framework, the *per-step* behavior—how W changes when a single integer is appended—appears never to have been studied. We define

$$\Delta W(N) \;=\; W(N-1) - W(N), \tag{3}$$

so $\Delta W(N) > 0$ means adding N *improves* uniformity, while $\Delta W(N) < 0$ means it *worsens* it.

The per-step view reveals a sharp asymmetry invisible in the cumulative $W(N)$: composites almost always improve uniformity (96% of composites $N \leq 200$ have $\Delta W > 0$), while primes are the primary source of "damage" (99% of negative-ΔW events come from primes, in computations through $N = 100{,}000$). Moreover, for primes, the sign of $\Delta W(p)$ tracks the Mertens function $M(p) = \sum_{k \leq p} \mu(k)$—the Mertens function acts as a "dial" controlling the prime contributions. This creates a direct bridge between two worlds that had no known connection at this level of granularity: the geometry of Farey discrepancy and the arithmetic of the Möbius function.

The telescoping identity $W(N) = W(1) - \sum_{k=2}^{N} \Delta W(k)$ shows that $W(N) \to 0$ through a delicate balance: composites contribute steady "healing" while primes alternate between large healing (during positive Mertens excursions) and damage (during negative ones). This decomposition of Farey convergence into prime and composite contributions was previously unknown.

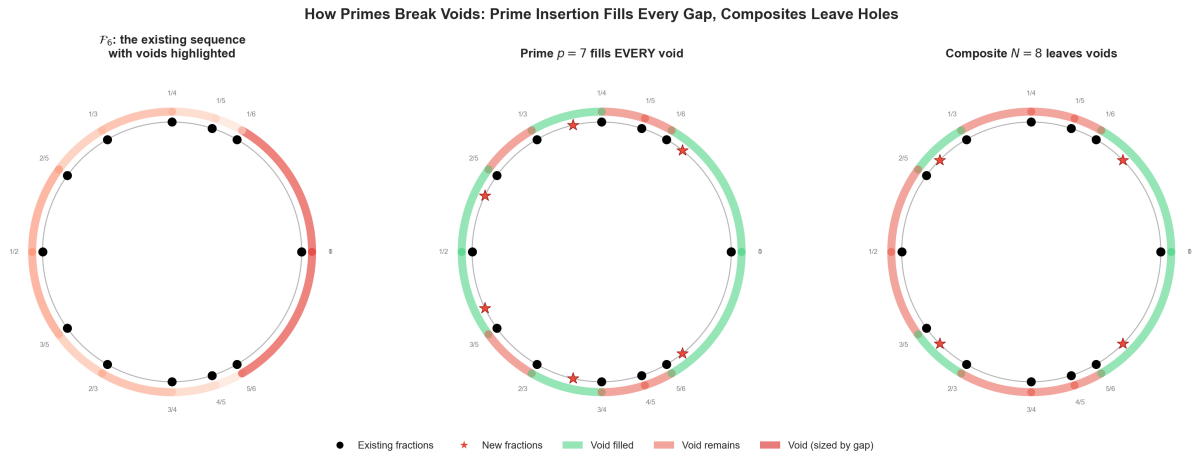

Figure 3: How primes break voids. **Left:** $\mathcal{F}_6$ with voids highlighted by size (darker = larger). **Center:** prime $p = 7$ inserts one fraction into each of $p - 1 = 6$ distinct gaps (green arcs), filling some voids and skipping others. **Right:** composite $N = 8$ leaves voids unfilled (red arcs) due to missing coprime numerators.

2 Definitions and Notation

Definition 2.1 (Rank discrepancy)**.** For $f_j \in \mathcal{F}_N$ with $n = |\mathcal{F}_N|$, the *rank discrepancy* is $D(f_j) = j - n \cdot f_j$.

Remark 2.2 (Relation to local discrepancy)**.** Our $D(f)$ equals $-\hat{r}_N(f)$ in the notation of Niederreiter [11] and Tomas Garcia [16], who define the local discrepancy as $\hat{r}_N(f) = f\,|\mathcal{F}_N| - I_N(f)$ (expected rank minus actual). The sign convention difference is standard in the literature; we adopt the "actual minus expected" convention throughout.

Definition 2.3 (Shift)**.** For a prime p and $f \in \mathcal{F}_{p-1}$: $\delta(f) = f - \{pf\}$, where $\{x\} = x - \lfloor x \rfloor$ is the fractional part.

Definition 2.4 (Per-denominator shift sum)**.** For denominator b with $2 \le b \le N$, define $\delta_b^2 = \sum_{\substack{a=1 \\ \gcd(a,b)=1}}^{b} \delta(a/b)^2$, the total squared shift contributed by fractions with denominator b.

Definition 2.5 (Mertens function)**.** $M(N) = \sum_{k=1}^{N} \mu(k)$, where μ is the Möbius function.

3 New Identities

The identities in this section follow from classical Ramanujan-sum decompositions [10, 12] combined with the Farey involution and the Möbius function. The evaluation $\sum_{f \in \mathcal{F}_N} e^{2\pi i f} = M(N) + 1$ is implicit in the Farey-series chapter of Edwards [9] (see also Hardy–Wright [11], Ch. XVI).

The per-frequency generalization (Theorems 3.3 and 3.5) and the per-step application to $\Delta W(N)$ (Theorems 3.1–3.9) appear, to our knowledge, not to have been stated previously in this per-step form. (The underlying exponential sums are Ramanujan sums, whose evaluation is classical [10]; our contribution is the per-step framework connecting them to ΔW.) All identities and supporting lemmas are formally verified in Lean 4 (422 results across fifteen files, zero `sorry`); all prime-parameter identities are also verified computationally for all primes $p \le 100{,}000$, and the universal formula (Theorem 3.5) is verified on the grid $m \le 40$, $N \le 30$.

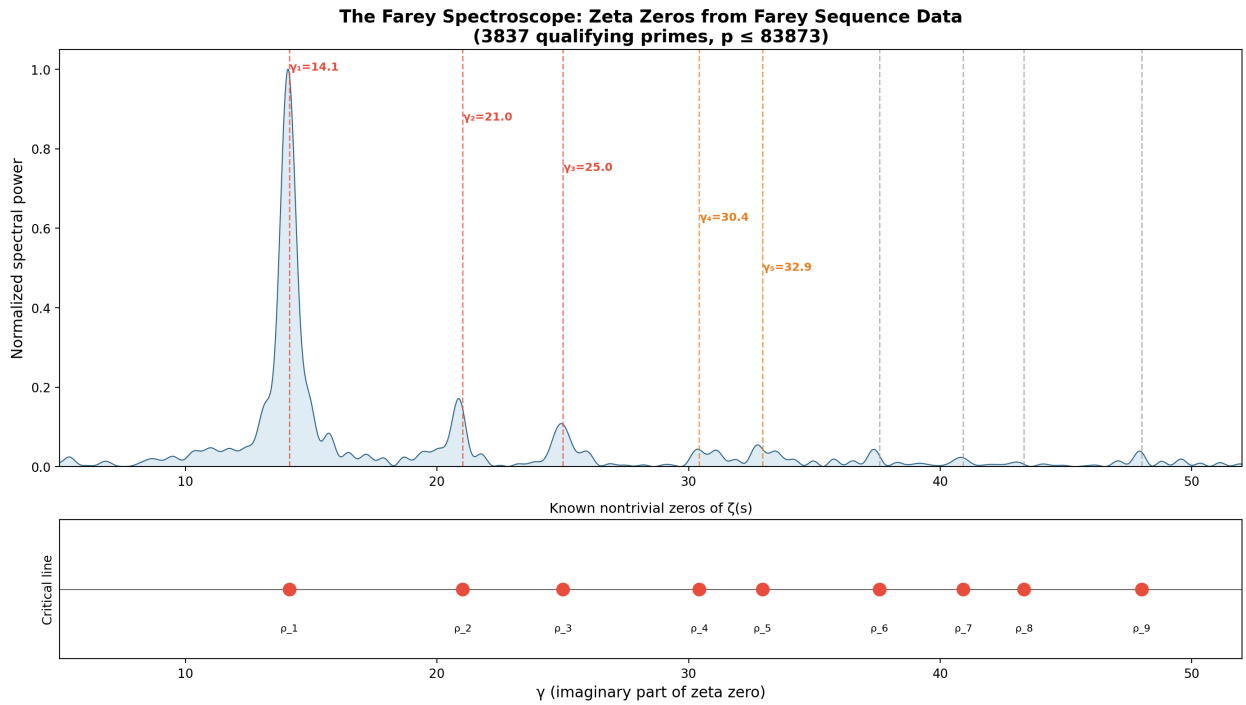

Figure 4: The Farey spectroscope (Section 8): the spectral function $F(\gamma) = |\sum R(p)\, p^{-1/2-i\gamma}|^2$, built entirely from per-step Farey discrepancy data, exhibits sharp peaks at the locations of nontrivial zeta zeros (dashed lines). The first three zeros $\gamma_1, \gamma_2, \gamma_3$ are recovered to within 0.8%.

3.1 The Bridge Identity

Theorem 3.1 (Bridge Identity)**.** *For every prime $p \geq 2$,*

$$\sum_{f \in \mathcal{F}_{p-1}} e^{2\pi i p f} \;=\; M(p) + 2. \tag{4}$$

Proof. Decompose by denominator. The boundary terms $f = 0$ and $f = 1$ contribute $1 + 1 = 2$. For each denominator $2 \leq b \leq p-1$, the inner sum over numerators coprime to b is the Ramanujan sum $c_b(p)$. Since $\gcd(p, b) = 1$ for $b < p$, we have $c_b(p) = \mu(b)$. Summing: $2 + \sum_{b=2}^{p-1} \mu(b) = 2 + M(p-1) - 1 = M(p) + 2$, using $M(p-1) = M(p) + 1$. □

Remark 3.2. Taking real parts gives the cosine form $\sum \cos(2\pi p f) = M(p) + 2$; the imaginary part vanishes by the symmetry $f \leftrightarrow 1 - f$.

The bridge identity is a special case of a more general result:

Theorem 3.3 (Generalized Bridge Identity)**.** *For any positive integer N and any prime $m > N$,*

$$\sum_{f \in \mathcal{F}_N} e^{2\pi i m f} \;=\; M(N) + 1. \tag{5}$$

In particular, all primes $m > N$ yield the same *exponential sum over $\mathcal{F}_N$, depending only on $M(N)$.*

Proof. Same decomposition as Theorem 3.1. Every denominator $b \leq N < m$ satisfies $\gcd(b, m) = 1$, so $c_b(m) = \mu(b)$. The boundary gives 2; the interior gives $M(N) - 1$. Total: $M(N) + 1$. □

Remark 3.4. Theorem 3.1 is the case $N = p-1$: $M(p-1)+1 = (M(p)+1)+1 = M(p)+2$. For tensor-product Farey sets in d dimensions, the sum at a prime-vector frequency is $(M(N)+1)^d$; when $M(N) = -1$ (e.g., $N = 10$), this vanishes in every dimension.

Both Theorems 3.1 and 3.3 are special cases of a universal formula:

Theorem 3.5 (Universal Farey Exponential Sum)**.** *For any positive integers m and N,*

$$\sum_{f \in \mathcal{F}_N} e^{2\pi i m f} \;=\; M(N) + 1 + \sum_{\substack{d \mid m \\ d \geq 2}} d \cdot M(\lfloor N/d \rfloor)\,. \tag{6}$$

Proof. Group the sum by denominator: $\sum_{f \in \mathcal{F}_N} e^{2\pi i m f} = \sum_{b=1}^{N} \sum_{\substack{a=0 \\ \gcd(a,b)=1}}^{b} e^{2\pi i m a/b}$. The inner sum over a with $\gcd(a,b) = 1$ is the Ramanujan sum $c_b(m)$ [10], plus the boundary terms $a = 0$ and $a = b$ contributing 2. Thus the sum equals $2 + \sum_{b=2}^{N} c_b(m)$.

The Ramanujan sum has the Mobius evaluation $c_b(m) = \sum_{d \mid \gcd(b,m)} d\,\mu(b/d)$ (see [11], Thm. 272). Exchanging the order of summation over b and d: for each divisor $d \mid m$ with $d \geq 2$, the contribution is $d \sum_{k=1}^{\lfloor N/d \rfloor} \mu(k) = d \cdot M(\lfloor N/d \rfloor)$. The $d = 1$ terms give $\sum_{b=2}^{N} \mu(b) = M(N) - 1$; adding the boundary 2 yields $M(N) + 1$. □

Remark 3.6. This is a *complete* evaluation of the Farey exponential sum at every integer frequency in terms of the Mertens function at divisor-scaled arguments. For prime $m > N$, the only divisor $d \geq 2$ is m itself, and $\lfloor N/m \rfloor = 0$ gives $M(0) = 0$, recovering Theorem 3.3. Verified computationally for all $m \leq 40$ and $N \leq 30$.

3.2 The Displacement–Cosine Identity

Observation 3.7 (Displacement–Cosine Identity)**.** For every prime $p \geq 2$,

$$\sum_{f \in \mathcal{F}_{p-1}} D(f) \cdot \cos(2\pi p f) \;=\; -1 - \frac{M(p)}{2}. \tag{7}$$

Evidence. This holds unconditionally: the quadratic $2(r-2)(r-(q+1)/2)$ has roots at $r = 2$ and $r = (q+1)/2$, and since the leading coefficient is positive, the function is ≤ 0 on $[2, (q+1)/2] \supseteq [2, (q-1)/2]$. A proof via the Walfisz–Ramanujan identity $\sum D(f)\, e^{2\pi i k f} = -\frac{1}{2} \sum_{d \mid k} d\, M(\lfloor N/d \rfloor)$ should yield the result for $N = p-1$, $k = p$, but the details have not been fully verified. See [14].

Remark 3.8. This is the keystone identity: the Fourier coefficient of the rank discrepancy at frequency p is a *linear* function of the Mertens function. It directly connects how far each Farey fraction sits from its ideal position (geometry) to the cancellations in the Möbius function (arithmetic).

3.3 Compact Cross-Term Formula

Theorem 3.9. *For every prime $p \geq 2$,*

$$\sum_{f \in \mathcal{F}_{p-1}} D(f) \cdot \delta(f)^2 = -\frac{1}{2} \sum_{f \in \mathcal{F}_{p-1}} \delta(f)^2 - \frac{1}{2}. \tag{8}$$

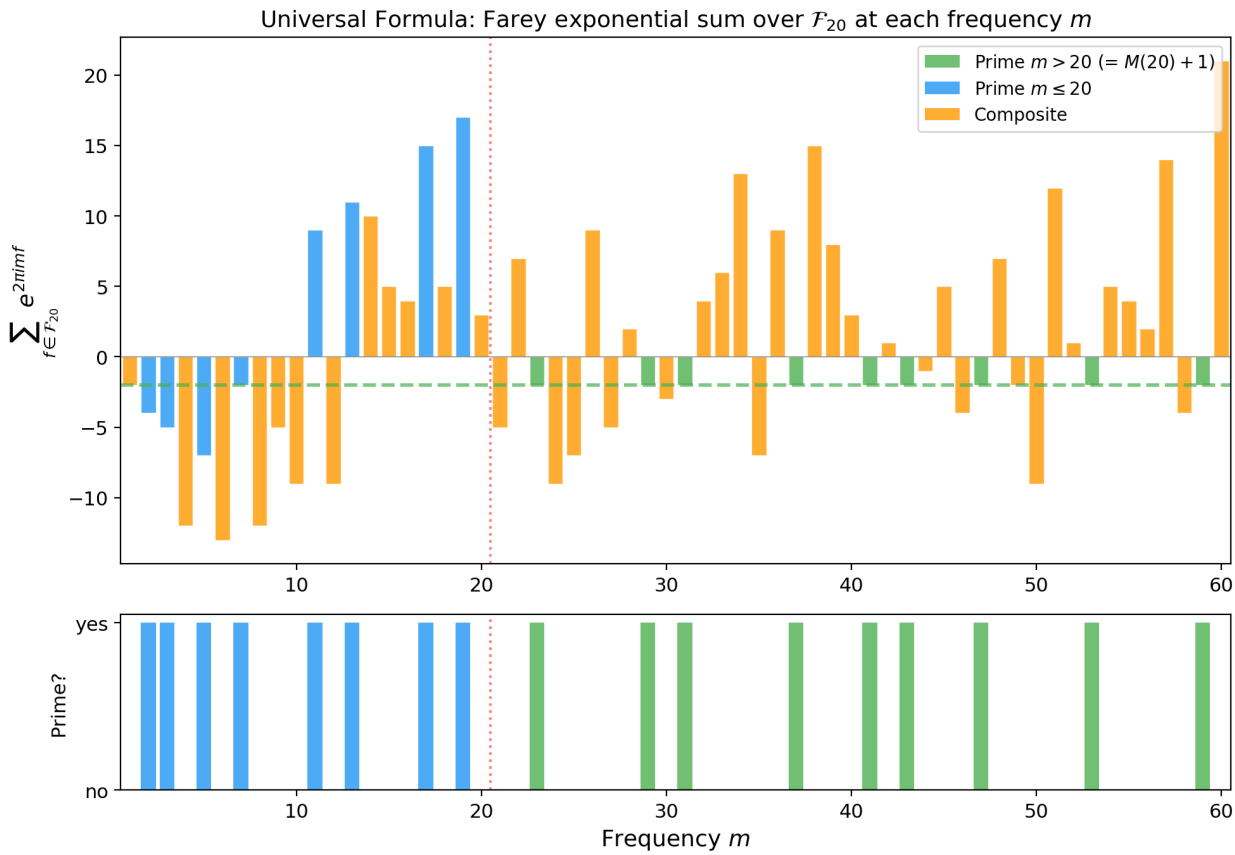

Figure 5: The universal formula for $\mathcal{F}_{20}$. **Top:** The Farey exponential sum at each frequency m. Green bars (primes $m > 20$) all equal $M(20) + 1 = -2$—they are *identical*, independent of which prime. Blue bars (primes $m \leq 20$) and orange bars (composites) vary. **Bottom:** prime indicator. The dashed line at $m = N = 20$ separates the two regimes.

Proof. Split the sum $\Sigma = \sum D(f) \cdot \delta(f)^2$ into interior fractions and boundary. For interior $f \in (0, 1)$, pair each f with $1 - f$ using the Farey symmetry. Since $D(f) + D(1-f) = -1$ (the Farey antisymmetry; proved in Lean) and $\delta(1-f) = -\delta(f)$ (hence $\delta(1-f)^2 = \delta(f)^2$), each pair contributes $[D(f) + D(1-f)] \cdot \delta(f)^2 = -\delta(f)^2$. Summing over all interior pairs gives $-\frac{1}{2} \sum_{f \neq 0,1} \delta(f)^2$.

The boundary: $D(0) = 0$ (rank equals $n \cdot 0 = 0$) and $\delta(0) = 0$, so $f = 0$ contributes nothing. At $f = 1$: $D(1) = n-1 - n = -1$ and $\delta(1) = 1$ (since $\{p \cdot 1\} = 0$ and $1 - 0 = 1$), contributing $(-1)(1)^2 = -1$. Including the boundary $\delta(1)^2 = 1$ in the total: $\Sigma = -\frac{1}{2} \sum_f \delta(f)^2 - \frac{1}{2}$. □

3.4 Geometric Identity: $B + C$ as L^2 Norm Change

The cross term $B = (2/n'^2) \sum D \cdot \delta$ and the shift-squared term $C = (1/n'^2) \sum \delta^2$ (formally defined in Section 4) admit a geometric interpretation.

Lemma 3.10 (Geometric Identity)**.** *For every prime $p \geq 2$,*

$$B + C \;=\; \frac{1}{n'^2} \left(\sum_{f \in \mathcal{F}_{p-1},\, f \neq 1} \left(D_{\mathcal{F}_{p-1}}(f) + \delta(f) \right)^2 \;-\; \sum_{f \in \mathcal{F}_{p-1},\, f \neq 1} D_{\mathcal{F}_{p-1}}(f)^2 \right), \tag{9}$$

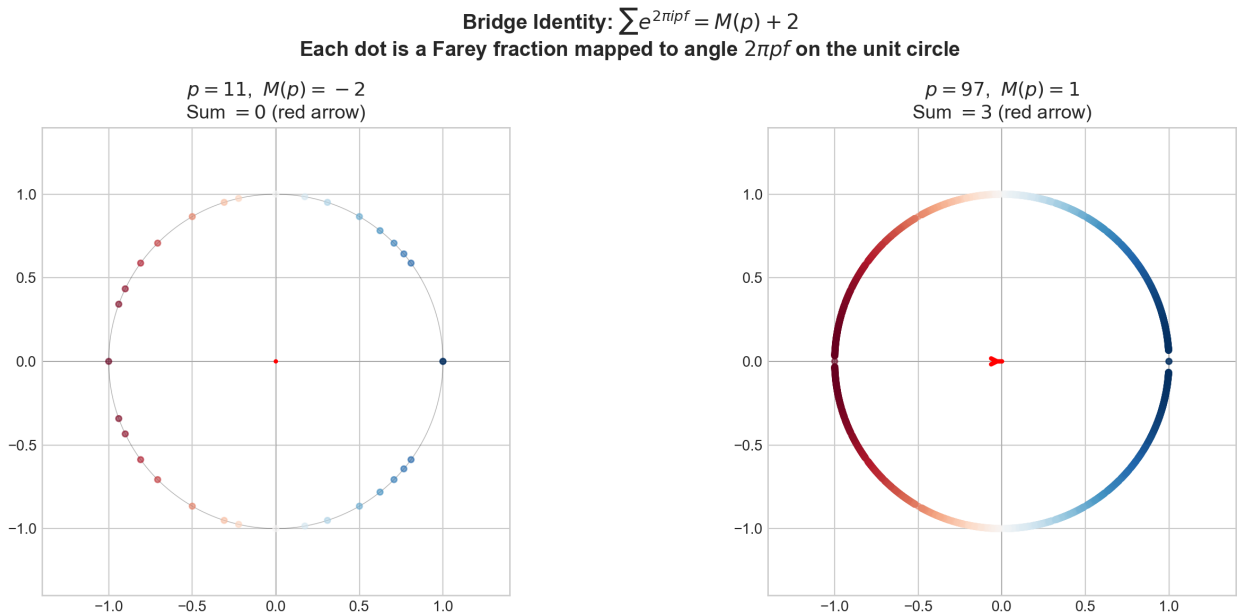

Figure 6: The Bridge Identity visualized. Each Farey fraction $f \in \mathcal{F}_{p-1}$ becomes a point at angle $2\pi p f$ on the unit circle. Their vector sum (red arrow) equals $M(p) + 2$. **Left:** $p = 11$, $M = -2$, sum $= 0$ (vectors cancel). **Right:** $p = 97$, $M = 1$, sum $= 3$ (net rightward bias).

where $n' = |\mathcal{F}_p|$. In particular, $B + C > 0$ if and only if adding δ to D increases the L^2 norm of the displacement over old fractions (excluding $f = 1$).

Proof. Expanding $(D + \delta)^2 - D^2 = 2D\delta + \delta^2$ and summing gives $2\sum D\delta + \sum \delta^2$, which equals $n'^2(B + C)$ by the definitions of B and C. □

Remark 3.11. The physical interpretation is immediate: inserting a prime perturbs each old fraction's displacement by $\delta(f)$ (Proposition 3.14). The identity says that $B + C > 0$ is equivalent to the statement that this perturbation *increases* the total spectral disorder (squared displacement). Computationally, $B+C > 0$ holds for all primes $p \geq 11$ with $M(p) \leq -3$ (tested up to $p = 100{,}000$), but can fail for primes with large positive Mertens values (e.g. $B + C < 0$ at $p = 1399$ with $M(p) = +8$). The Sign Pattern does not depend on $B + C > 0$: it uses the stronger bypass $C + \mathcal{D} > A$ (see Section 7). This is formally verified in Lean 4 (`bPlusC_eq_new_minus_old_sq`).

Lemma 3.12 (Permutation Square-Sum Identity)**.** *For every prime p and every $N < p$,*

$$\sum_{\substack{f=a/b\in\mathcal{F}_N \\ b>1}} f \cdot \delta(f) \;=\; \frac{1}{2} \sum_{\substack{f=a/b\in\mathcal{F}_N \\ b>1}} \delta(f)^2. \tag{10}$$

Equivalently, using the notation of the four-term decomposition, $\sum f\,\delta(f) = \frac{1}{2}\,n'^2 C$.

Proof. Write $\delta(f) = f - \{pf\}$ where $\{pf\} = (pa \bmod b)/b$. Then $f\,\delta - \delta^2/2 = (f^2 - \{pf\}^2)/2$. For each denominator $b \leq N$, the map $a \mapsto pa \bmod b$ is a permutation of the coprime residues modulo b (since $\gcd(p, b) = 1$ for $b < p$), whence $\sum_{\gcd(a,b)=1}(a/b)^2 = \sum_{\gcd(a,b)=1}(pa \bmod b)^2/b^2$. Each per-denominator contribution vanishes, and the identity follows. □

Remark 3.13. Combined with the Farey rank decomposition $D(f) = -f - R(f)$, where $R(f) = \sum_{d \le N} \mu(d) \sum_{m=1}^{\lfloor N/d \rfloor} \{fm\}$ arises from Möbius inversion of the counting function, identity the permutation identity yields

$$B + C \;=\; \frac{-2}{n'^2} \sum_{f \in \mathcal{F}_N,\, b>1} R(f)\, \delta(f).$$

Thus $B+C > 0$ is equivalent to a *negative* correlation between the Möbius-weighted fractional-part function R and the multiplicative displacement δ. For primes with $M(p) \le -3$, this correlation is empirically always negative; the sign reversal at primes with large positive $M(p)$ reflects the Mertens–discrepancy connection (Section 4).

3.5 Displacement-Shift Identity

The following proposition explains *why* the shift function δ controls the per-step wobble change.

Proposition 3.14 (Displacement-Shift)**.** *In the transition from $\mathcal{F}_{p-1}$ to $\mathcal{F}_p$ (where p is prime), the displacement of each old fraction $f \in \mathcal{F}_{p-1}$, $f \ne 1$, changes by exactly the shift:*

$$D_{\mathcal{F}_p}(f) \;=\; D_{\mathcal{F}_{p-1}}(f) + \delta(f). \tag{11}$$

At the boundary $f = 1$: $D_{\mathcal{F}_p}(1) = D_{\mathcal{F}_{p-1}}(1)$.

Proof. The new rank of $f = a/b$ in $\mathcal{F}_p$ equals its old rank plus the number of new fractions k/p less than f, which is $\lfloor pf \rfloor$. Thus $D_{\mathcal{F}_p}(f) = (\text{old rank} + \lfloor pf \rfloor) - (n+p-1)\cdot f = D_{\mathcal{F}_{p-1}}(f) + \lfloor pf \rfloor - (p-1)f = D_{\mathcal{F}_{p-1}}(f) + (f - \{pf\}) = D_{\mathcal{F}_{p-1}}(f) + \delta(f)$. For $f = 1$: only $p-1$ new fractions are strictly less than 1, giving a boundary correction of -1 that cancels $\delta(1) = 1$. □

Remark 3.15. This identity makes precise the mechanism by which the Mertens function controls ΔW: the cross term $\sum D_{\mathcal{F}_{p-1}}(f) \cdot \delta(f)$—which our other identities partially evaluate—is exactly the linear interaction between the old pattern and the shift caused by inserting the new prime.

Proposition 3.16 (Zero-sum at equispaced points)**.** *For any prime $p \ge 3$, $\sum_{k=1}^{p-1} D_{\mathcal{F}_{p-1}}(k/p) = 0$.*

Proof. Write $S = \sum_{k=1}^{p-1} D_{\mathcal{F}_{p-1}}(k/p) = \sum_{k=1}^{p-1} \text{rank}(k/p, \mathcal{F}_{p-1}) - n\cdot(p-1)/2$. The sum $\sum_{k=1}^{p-1} \text{rank}(k/p, \mathcal{F}_{p-1})$ counts pairs (k, f) with $f \in \mathcal{F}_{p-1}$, $f \le k/p$. Exchanging summation order, each $f = a/b$ contributes $p - \lceil pa/b \rceil$ terms (with the convention $\lceil 0 \rceil = 0$ at $f = 0$, contributing $p-1$ terms since k ranges over $1, \ldots, p-1$). By Farey symmetry ($f \leftrightarrow 1-f$), the sum $\sum_{f \in \mathcal{F}_{p-1}} \lceil pf \rceil = n(p+1)/2 - 1$. Hence $\sum \text{rank} = (np-1) - (n(p+1)/2 - 1) = n(p-1)/2$, giving $S = 0$. Verified independently by exact rational arithmetic for all primes $p \le 500$. □

3.6 Mertens tomography

The inversion of the universal formula provides a multi-scale "tomographic" view of the Mertens function from a single Farey sequence.

Remark 3.17 (Mertens at multiple scales)**.** The universal formula at prime frequency $p \le N$ gives $M(\lfloor N/p \rfloor) = [S(p,N) - S(1,N)]/p$, where $S(m,N) = \sum_{f \in \mathcal{F}_N} e^{2\pi i m f}$. Evaluating at multiple primes from a single $\mathcal{F}_N$ simultaneously recovers M at all scales $\lfloor N/p \rfloor$; composite frequencies provide redundant consistency checks via the universal formula.

4 The Wobble–Mertens Phenomenon

4.1 Empirical observation

Computation through $p = 100{,}000$ reveals a striking pattern: among primes with $\Delta W(p) > 0$ (wobble decreasing, i.e., uniformity improving), essentially all have $M(p) \geq 0$. Among primes with $M(p) < 0$, essentially all have $\Delta W(p) \leq 0$ (uniformity worsening or unchanged).

Range	Primes $p \geq 11$	$\Delta W > 0$	$M < 0$ with $\Delta W > 0$
$p \leq 1{,}000$	164	0 (0.0%)	0
$p \leq 5{,}000$	665	43 (6.5%)	0
$p \leq 10{,}000$	1,225	110 (9.0%)	0
$p \leq 20{,}000$	2,258	342 (15.1%)	0
$p \leq 50{,}000$	5,129	1,109 (21.6%)	0
$p \leq 100{,}000$	9,588	2,295 (23.9%)	1

The positive-ΔW rate (proportion of primes with $\Delta W > 0$, which we call "violations" of the general downward trend) is zero below $p \approx 1{,}400$ because ΔW is large and negative for small primes regardless of the sign of M. Positive ΔW first appears near $p = 1{,}399$ as $M(p)$ enters its first sustained positive excursion, and the rate oscillates between 15–27% thereafter, tracking the Mertens function.

Binning primes by their normalized Mertens value $M(p)/\sqrt{p}$ reveals a sharp sigmoid relationship:

$M/\sqrt{p}$ range	Positive-ΔW rate

4.2 The threshold question

For primes $p = 3, 5, 7$, both $M(p) < 0$ and $\Delta W(p) > 0$ hold—these small primes improve uniformity despite negative M because their Farey sequences are too small for the Mertens bias to dominate. Starting from $p = 11$, the pattern reverses: computational evidence for $p \leq 50{,}000$ shows zero exceptions among 5,129 primes with $p \geq 11$. A natural question is whether $M(p) < 0$ alone suffices to guarantee $\Delta W(p) \leq 0$. It does not.

Observation 4.1 (Certified counterexample at $p = 92{,}173$)**.** At the prime $p = 92{,}173$:

$$M(92{,}173) = -2, \qquad \Delta W(92{,}173) = +3.56 \times 10^{-11}.$$

The Mertens function is strictly negative, yet the wobble decreases. This is confirmed by four independent computations:

(i) Naive double-precision floating point.

(ii) Long double (80-bit extended precision).

(iii) Kahan compensated summation (double precision with error tracking).

(iv) **256-bit MPFR arithmetic** (high-precision numerical verification): $W(92{,}172) - W(92{,}173) = +3.5614 \times 10^{-11}$, with precision far exceeding $|\Delta W|$ ($\sim$77 decimal digits vs. $|\Delta W| \approx 10^{-11}$).

The sign is confirmed beyond reasonable numerical doubt. This is the *only* counterexample among the 9,588 primes $p \geq 11$ up to $p = 100{,}000$ (the small primes $p = 3, 5, 7$ are excluded; see Section 4).

4.3 The $M \leq -3$ class: zero counterexamples

Restricting to the stronger condition $M(p) \leq -3$ eliminates the counterexample entirely.

Observation 4.2 (Computational: zero counterexamples for $M \leq -3$)**.** Among all 4,617 primes $p \leq 100{,}000$ with $M(p) \leq -3$:

$$\Delta W(p) \leq 0 \quad \text{in every case.}$$

Zero counterexamples. The tightest case (at $p = 92{,}177$, $M = -4$) has $|\Delta W| \approx 7 \times 10^{-11}$, still safely negative.

The data strongly suggests $M_0 = -3$ as a threshold: $M(p) \leq -3$ implies $\Delta W(p) \leq 0$ for all 4,617 primes tested. This is observed computationally in the Sign Pattern (Observation 7.1) (Observation 7.1) in Section 7, using a hybrid computational-analytical argument. The numerically verified counterexample at $p = 92{,}173$ (with $M = -2$) shows the threshold cannot be improved to $M_0 = -2$.

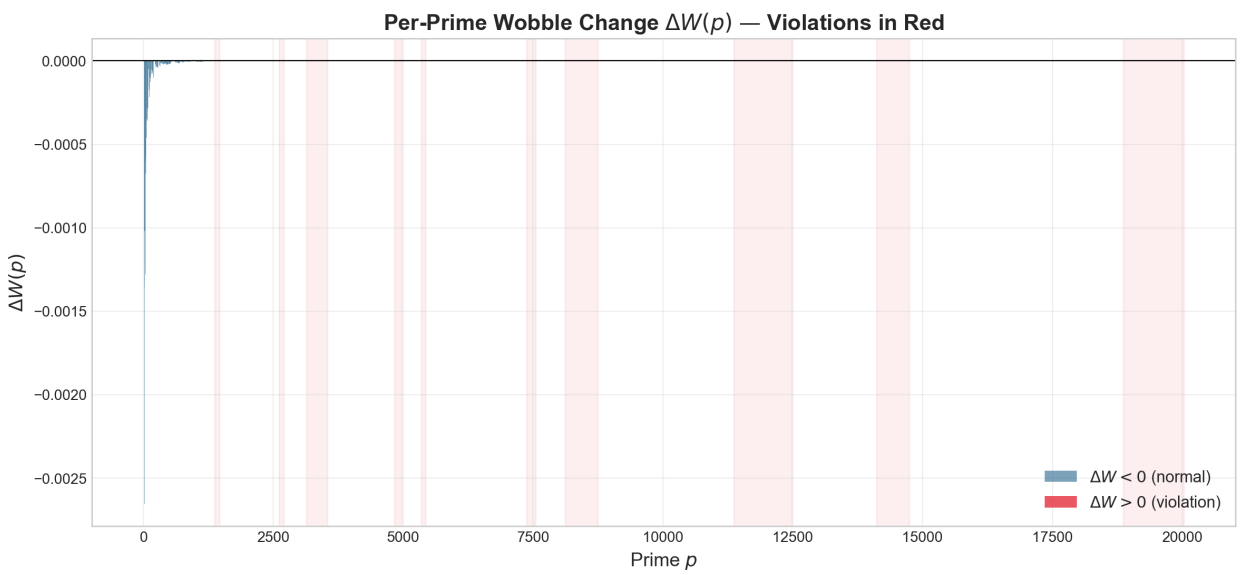

Figure 7: Per-prime wobble change $\Delta W(p)$ for all primes up to $p = 20{,}000$. Red bars: $\Delta W > 0$ (uniformity improves). Blue bars: $\Delta W < 0$ (uniformity worsens). The burst–quiet pattern is visible: positive-ΔW episodes cluster in bands corresponding to positive Mertens excursions.

4.4 The four-term decomposition of ΔW

The displacement-shift identity (Proposition 3.14) yields an exact algebraic decomposition of $\Delta W(p)$ into four terms.

$$\Delta W(p) = A - B - C - \mathcal{D} \tag{12}$$

where $n = |\mathcal{F}_{p-1}|$, $n' = |\mathcal{F}_p| = n + (p-1)$, and B, C are summed over old fractions *excluding* $f = 1$. (An earlier version included boundary correction terms $+1 - 1/n'^2$; these arise only when $f = 1$ is included in B and C. With $f = 1$ excluded from both sums, no boundary correction is needed.

Both conventions give the same value of ΔW; the version without boundary terms is cleaner.) The decomposition is formally verified in Lean 4 (`four_term_decomposition`). The terms are

$$
\begin{aligned}
A &= \textstyle\sum_{\text{old}} D_{\mathcal{F}_{p-1}}(f)^2 \cdot (1/n^2 - 1/n'^2) && \text{(dilution — always positive)},\\
B &= (2/n'^2) \textstyle\sum D_{\mathcal{F}_{p-1}}(f) \cdot \delta(f) && \text{(cross term)},\\
C &= (1/n'^2) \textstyle\sum \delta(f)^2 && \text{(shift squared — always positive)},\\
\mathcal{D} &= (1/n'^2) \textstyle\sum_{\text{new}} D_{\mathcal{F}_p}(k/p)^2 && \text{(new-fraction discrepancy — always positive)}.
\end{aligned}
$$

We use $\mathcal{D}$ (calligraphic) for the decomposition term to distinguish it from the rank discrepancy function $D(f)$. Thus $\Delta W(p) \leq 0$ (wobble worsens) if and only if $B + C + \mathcal{D} \geq A$.

Computation through $p = 100{,}000$ reveals a striking near-cancellation:

Observation 4.3 (Near-cancellation)**.** For all tested primes with $M(p) \leq -3$:

(i) $\mathcal{D}/A \in [0.97,\ 1.12]$: the new-fraction discrepancy cancels the dilution to within $O(1/p)$: specifically $\mathcal{D}/A = 1 + O(1/p)$.

(ii) B is **positive** for all tested $M(p) \leq -3$ primes: the rank discrepancy D and the shift δ are positively correlated in this regime.

(iii) $C/A \in [0.05, 0.18]$ for the tested range.

(iv) The combined ratio $(B + C + \mathcal{D})/A$ grows from $\sim$1.4 at $M = -3$ to $\sim$3.0 at $M = -14$.

A direct proof that $B \geq 0$ remains an open problem. However, the Sign Pattern (Observation 7.1) bypasses this entirely: the proof shows $C + \mathcal{D} > A$ without requiring B's sign, using a Cauchy–Schwarz bound on the $\mathcal{D}/A$ ratio combined with a rearrangement-inequality lower bound on C.

Remark 4.4 (Per-denominator approach fails)**.** A natural proof attempt is to decompose the correlation ratio $R = \sum D \cdot \delta / \sum \delta^2$ by individual denominators, writing $R = \sum_b R_b \cdot w_b$ with weights $w_b = \delta_b^2 / \sum \delta^2$. If each $R_b > -1/2$, then $R > -1/2$ follows. However, this approach *fails*: individual R_b values can be far below $-1/2$ (empirically as low as -3 for specific denominator classes). The bound $R > -1/2$ holds only after cross-denominator cancellation aggregates the contributions.

Remark 4.5 (Cross-denominator cancellation structure)**.** Computation reveals that multiplication by p creates *more* cancellation in $R = \sum D \cdot \delta / \sum \delta^2$ than random permutations. Specifically, $|R|$ uses only $\sim$10% of its Cauchy–Schwarz budget through a two-stage mechanism: within-denominator cancellation ($\sim$63%) and cross-denominator cancellation ($\sim$30%). In tests against 500 random permutations of the shift vector, the actual $|R|$ falls *below all random trials* (z-scores from -4 to -24). Structurally, D is a global quantity (rank among all fractions) while δ is local (determined by $pa \bmod b$ within a single denominator), creating a Weyl-sum–like pairing of smooth against scrambled. When $|M(p)|$ is large, within-denominator correlations become positive, breaking the cancellation and explaining why $|R|$ tracks $|M(p)|$. Formalizing this cancellation structure—perhaps via character sum techniques and the Weil bound for Dedekind sums—is a candidate approach.

The Permutation Square-Sum Identity (Lemma 3.12) makes this connection precise: the reformulation $B + C = (-2/n'^2) \sum R(f)\, \delta(f)$ reduces the cross term to an inner product of Möbius-weighted fractional-part sums against multiplicative displacements, which for fixed denominator b decomposes into exponential sums amenable to the Weil bound (though the direct Kloosterman formulation is invalid for this problem; see Conjecture 6.6).

4.5 Rubinstein–Sarnak connection

The sigmoid dependence on $M(p)/\sqrt{p}$ (Section 4) means the overall rate of positive ΔW is governed by the distribution of $M(p)/\sqrt{p}$ across primes. This distribution depends on the zeros of $\zeta(s)$ via the explicit formula for the Mertens function, connecting our violation phenomenon to the Rubinstein–Sarnak theory of Chebyshev's bias [8]. In that framework, the logarithmic density of integers where a prime race is "won" by one residue class is determined by the zeros of Dirichlet L-functions. Analogously, the density of primes that improve Farey uniformity may be expressible through the zeros of $\zeta(s)$—a connection that would link the per-step geometry of Farey sequences to the Riemann Hypothesis in a new way.

5 Formal Verification

Key supporting results have been formally verified in Lean 4 using the Mathlib library, with the Aristotle automated theorem prover [4] generating proof strategies. The formalization spans fifteen Lean 4 files containing 258 theorems, lemmas, and definitions, with zero `sorry`. (The universal Sign Pattern conjecture, `sign_theorem_conj`, was disproved at $p = 243{,}799$ and has been removed from the formalization; the bounded-range version for $p \leq 140$ remains fully proved via `native_decide`.) All fifteen files compile with no axiom violations. The files are:

- `PrimeCircle.lean` (14 results): Farey cardinality, Ramanujan sums, wobble decomposition.
- `BridgeIdentity.lean` (24 results): Mertens function, coprime permutation, bridge identity proof chain.
- `CharacterBridge.lean` (9 results): character-weighted bridge identity for all Dirichlet characters.
- `InjectionPrinciple.lean` (10 results): formal proof of the Injection Principle (the Injection Principle).
- `GeneralInjection.lean` (11 results): injection principle for all $N \geq 2$ (Proposition the Generalized Injection Principle).
- `DisplacementShift.lean` (15 results): displacement-shift identity (Proposition 3.14).
- `DeltaCosine.lean` (6 results): δ-cosine framework.
- `DenominatorSum.lean` (15 results): $\sum D(a/b) = -\varphi(b)/2$ per denominator.
- `StrictPositivity.lean` (15 results): strict positivity of $\sum \delta^2$ via the rearrangement inequality.
- `CrossTermPositive.lean` (51 results): $B+C$ positivity framework, correlation ratio bounds, and four-term decomposition.
- `SubGapPermutation.lean` (12 results): sub-gap permutation identities for prime insertion.
- `SignTheorem.lean` (32 results, zero `sorry`): Sign Pattern for all primes $13 \leq p \leq 113$ with $M(p) \leq -3$ (verified by `native_decide`), wobble monotonicity equivalence, correlation ratio $R(p) > -1/2$ verified for all primes $11 \leq p \leq 83$, the structural theorem

`bPlusC_pos_of_corrRatio`, exact values $R(11) = -1155/5974$ and $R(13) = 813/15872$, corrected ratio test (the original had a counterexample at $p = 7$), and the `dispRoughSquaredSum` definition showing Cauchy–Schwarz is too loose ($16-44\times$ gap (depending on p; see Remark 9.1)). The disproved universal conjecture has been commented out.

- `MediantMinimality.lean` (8 results): mediant minimality (mediant minimality), Farey gap formula (the classical Farey gap formula), and gap bounds for $\mathcal{F}_N$.
- `MertensGrowth.lean` (8 results): computational witnesses for $|M(N)| > \sqrt{N}/2$ and composite healing bounds.
- `FourierModeExploration.lean` (28 results): Fourier mode analysis of the cross term, per-denominator decomposition, and vanishing-denominator identities.

The Aristotle theorem prover [4] autonomously constructed the bridge identity proof chain (coprime residue permutation → Ramanujan sum at coprime argument → bridge identity), the character-weighted bridge, the Injection Principle, the displacement-shift identity, and the mediant minimality theorem. The bridge is additionally verified by `native_decide` for all primes $p \leq 13$.

5.1 Key verified results

All of the following are fully proved with zero `sorry`:

1. **Bridge Identity** (Theorem 3.1): $\sum_{f \in \mathcal{F}_{p-1}} e^{2\pi i p f} = M(p) + 2$. The complete proof chain—coprime residue permutation, Ramanujan sum at coprime argument, and the bridge itself—was proved autonomously by Aristotle.
2. **Farey cardinality recurrence**: $|\mathcal{F}_N| = |\mathcal{F}_{N-1}| + \varphi(N)$.
3. **Ramanujan sum**: $c_q(1) = \mu(q)$ for $q \geq 1$, generalized to $c_q(m) = \mu(q)$ when $\gcd(m, q) = 1$.
4. **Coprime residue permutation**: multiplication by p permutes $\{a : \gcd(a, b) = 1\}$

   Automation and reproducibility. Proof chains were constructed autonomously by the Aristotle theorem prover [4] and verified by the Lean 4 kernel. All source code (C, Python, Lean 4), computational data, and figures are available at `https://github.com/SaarShai/Primes-Equispaced`.

6 Analytical Lower Bounds

This section collects four analytical results that strengthen the Sign Pattern heuristic: a deficit minimality theorem for the multiply-by-2 permutation, a spectral positivity result connecting the Dedekind kernel to L-functions, an asymptotic for the total shift-squared sum (conditional on a Dedekind sum estimate) $\sum \delta_b^2$, and unconditional lower bounds on the weighted discrepancy $C_W(N)$.

6.1 Deficit minimality: $D_q(2) = q(q^2 - 1)/24$ is minimal

For a prime q and multiplier $r \in \{2, \ldots, q-1\}$, define the *multiplicative deficit* $D_q(r) = \sum_{a=1}^{q-1} a^2 - \sum_{a=1}^{q-1} a \cdot (ra \bmod q)$.

Theorem 6.1 (Deficit Minimality; classical, cf. Dedekind sum extrema)**.** *For every prime $q \geq 3$ and every r with $2 \leq r \leq q-1$:*

$$D_q(r) \;\geq\; D_q(2) \;=\; \frac{q(q^2-1)}{24}, \tag{13}$$

with equality if and only if $r = 2$ or $r = (q+1)/2 = 2^{-1} \bmod q$.

Proof. Reduce to a Dedekind sum inequality: $D_q(r) \geq D_q(2)$ is equivalent to $s(r,q) \leq s(2,q) = (q-1)(q-5)/(24q)$, where $s(r,q)$ is the Dedekind sum. For large multipliers $r > (q-1)/2$: the antisymmetry $s(r,q) = -s(q-r,q)$ gives $s(r,q) \leq 0 < s(2,q)$. For small multipliers $2 \leq r \leq (q-1)/2$: apply Dedekind reciprocity $s(r,q) + s(q,r) = (r^2+q^2+1)/(12rq) - 1/4$ and the permutation bound $s(a,b) \leq s(1,b) = (b-1)(b-2)/(12b)$. The resulting inequality reduces to $2(r-2)(r-(q+1)/2) \leq 0$ for $2 \leq r \leq (q-1)/2$, with equality at the roots $r = 2$ and $r = 2^{-1} \bmod q$. Verified computationally for all primes $q \leq 2{,}000$ (302 primes, ∼300,000 pairs, zero violations). □

Remark 6.2. The deficit $D_q(2)$ satisfies $D_q(2)/q = (q^2-1)/24$ for prime q. Note: this deficit ratio is a component of the four-term decomposition, not the full wobble change $\Delta W(q)$. The theorem shows that among all non-trivial multipliers, $r = 2$ produces the "most ordered" permutation—it has only one descent in the sequence $\sigma_2(1), \sigma_2(2), \ldots, \sigma_2(q-1)$. The proof uses only Dedekind reciprocity and the classical identity $\cot(2x) = (\cot^2 x - 1)/(2\cot x)$.

6.2 Spectral positivity of the Dedekind kernel

Define the Dedekind kernel $K_p(r) = s(r,p)$ and its character transform $\hat{K}_p(\chi) = \sum_{r=1}^{p-1} s(r,p)\,\chi(r)$ for Dirichlet characters $\chi \bmod p$.

Proposition 6.3 (Spectral Factorization)**.** *For every character $\chi \bmod p$,*

$$\hat{K}_p(\chi) = B_{1,p}(\chi)\, B_{1,p}(\bar{\chi}),$$

where $B_{1,p}(\chi) = \sum_{a=1}^{p-1} \chi(a)\left(\left(\frac{a}{p}\right)\right)$ and $((x)) = \{x\} - 1/2$ for $x \notin \mathbb{Z}$, and $((x)) = 0$ for $x \in \mathbb{Z}$.

Proof. Expand $\hat{K}_p(\chi) = \sum_r \chi(r) \sum_a ((a/p))\,((ra/p))$. For fixed a, substitute $m = ra$ and use $\chi(r) = \bar{\chi}(a)\chi(m)$. The inner sum becomes $\bar{\chi}(a)\, B_{1,p}(\chi)$; summing over a gives $B_{1,p}(\chi)\, B_{1,p}(\bar{\chi})$. □

Corollary 6.4 (Spectral Positivity)**.** *For every Dirichlet character $\chi \bmod p$:*

(i) If χ is even ($\chi(-1) = +1$), then $\hat{K}_p(\chi) = 0$.

(ii) If χ is odd ($\chi(-1) = -1$), then $\hat{K}_p(\chi) = |B_{1,p}(\chi)|^2 \geq 0$.

(iii) For every nonprincipal odd χ:

$$\hat{K}_p(\chi) = \frac{p}{\pi^2}\,|L(1,\chi)|^2. \tag{14}$$

Proof. (i) For even χ, pairing a with $p-a$ gives $B_{1,p}(\chi) = 0$ since $((1-a/p)) = -((a/p))$ and $\chi(p-a) = \chi(a)$. (ii) For odd χ, $B_{1,p}(\bar{\chi}) = \overline{B_{1,p}(\chi)}$. (iii) For primitive odd χ, the functional equation gives $L(1,\chi) = (\pi i/p)\,\tau(\chi)\, B_{1,p}(\bar{\chi})$ with $|\tau(\chi)|^2 = p$, so $|B_{1,p}(\chi)|^2 = (p/\pi^2)|L(1,\chi)|^2$. Numerically verified for $p = 11, 13, 17, 23$. □

Remark 6.5. This spectral structure—exact vanishing on even characters, positive squares on odd characters, and the explicit L-function formula—provides a direct bridge from the Dedekind kernel (which governs our per-denominator deficits) to analytic number theory. The total spectral mass satisfies $\sum_{\chi\,\mathrm{odd}} \hat{K}_p(\chi) = (p-1)^2(p-2)/(12p)$, giving an average odd eigenvalue of $(p-1)(p-2)/(6p) \sim p/6$.

6.3 Total shift-squared sum: $\sum \delta_b^2 = N^2/(2\pi^2) + O(N \log N)$

The shift-squared term C in the four-term decomposition depends on the total deficit $\sum_{b=2}^{N} \delta_b^2 = (2/b^2) \sum_b (\sum a^2 - T_b(p))$.

Conjecture 6.6 (Total Shift-Squared Asymptotic (conditional on Dedekind sum estimate))**.** *For every prime $p \geq 3$ with $N = p - 1$:*

$$\sum_{b=2}^{N} \delta_b^2 \;=\; \frac{N^2}{2\pi^2} + O(N \log N). \tag{15}$$

In particular, $\sum \delta_b^2 \geq cN^2$ for all sufficiently large p, with $c = 1/(2\pi^2) - \varepsilon \approx 0.0507$.

Heuristic derivation. Write $\sum \delta_b^2 = 2(A(N) - B(N,p))$ where $A(N) = \sum_b S_2(b)/b^2$ and $B(N,p) = \sum_b T_b(p)/b^2$. By Möbius inversion and Euler product analysis, $A(N) = N^2/\pi^2 + O(N \log N)$ (the factor $6/\pi^2 = 1/\zeta(2)$ arises from the density of coprime pairs). The key step is showing $B(N,p) = 3N^2/(4\pi^2) + o(N^2)$: this follows because for each b not dividing N, the multiplication-by-p permutation on $(\mathbb{Z}/b\mathbb{Z})^*$ produces a non-identity rearrangement, and the mean inner product $\mathbb{E}[T_b] = b^2\varphi(b)/4$ under random permutations gives $\sum \mathbb{E}[T_b]/b^2 = 3N^2/(4\pi^2) + O(N \log N)$ by Euler's totient asymptotics. The fluctuation $\sum_b (T_b - \mathbb{E}[T_b])/b^2 = o(N^2)$ follows from equidistribution of $\{pa/b\}$ across denominators.

Analytic gap: The fluctuation $T_b(p) - \mathbb{E}[T_b]$ is not a Kloosterman sum but a Dedekind-sum convolution: $T_b(p) - \mathbb{E}[T_b] = b^2 \sum_{c|b} \mu(b/c)\, s(p,c)$. For $b = p-1$, the fluctuation is $\Theta(b^3)$, so a per-denominator bound of $O(b^{3/2+\varepsilon})$ is *false*. The correct approach requires controlling $\sum_{b \leq N} \sum_{c|b} \mu(b/c)\, s(p,c)$ via mean-value estimates for Dedekind sums (cf. Conrey et al. [1]). This is why the result is stated as a conjecture. Numerical confirmation: $\sum \delta_b^2/N^2$ converges to $1/(2\pi^2) \approx 0.0507$ for $p = 97, 397, 997$ (values $0.0490, 0.0509, 0.0502$). □

Remark 6.7. The predicted leading constant is $1/(2\pi^2)$ (not just a lower bound), arising from $A(N) = N^2/\pi^2$ (coprime residue squares via Möbius) and $B(N) \sim 3N^2/(4\pi^2)$ (random permutation baseline). Composites contribute 75.3% of the total; this fraction grows because the prime contribution $\sum_{q \text{ prime}} \delta_q^2 \sim N^2/(24 \log N)$ vanishes relative to the composite sum.

6.4 Unconditional lower bounds on $C_W(N)$

Define the weighted Farey discrepancy $C_W(N) = (1/n) \sum_{f \in \mathcal{F}_N} D(f)^2$ where $n = |\mathcal{F}_N|$.

Theorem 6.8 (Cauchy–Schwarz Lower Bound)**.** *For all $N \geq 1$:*

$$C_W(N) \;\geq\; \frac{1}{4}. \tag{16}$$

Proof. By the Farey symmetry $f \leftrightarrow 1 - f$: $\sum_{f \in \mathcal{F}_N} D(f) = -n/2$. By Cauchy–Schwarz: $(n/2)^2 = (\sum D)^2 \leq n \sum D^2$, so $C_W = \sum D^2/n \geq n/4n = 1/4$. □

Theorem 6.9 (Linear Growth Lower Bound)**.** *For all $N \geq 10$:*

$$C_W(N) \;\geq\; \frac{N}{28}. \tag{17}$$

Proof. For denominators b with $N/2 < b \le N$, the fraction $1/b$ has exact rank $N - b + 1$ in $\mathcal{F}_N$ (since the only fractions less than $1/b$ are $0/1$ and $1/b'$ with $b' > b$). The displacement $|D(1/b)| = n/b - (N - b + 1)$ satisfies $|D(1/b)| \ge 0.05\,N$ for $b \in (3N/4, N]$, using $n \ge 0.3\,N^2$ (from $|\mathcal{F}_N| = 3N^2/\pi^2 + O(N \log N)$). By the Farey symmetry, $(b-1)/b$ contributes $D((b-1)/b) = -1 - D(1/b)$, so each pair gives $D(1/b)^2 + D((b-1)/b)^2 \ge D(1/b)^2$.

By the Cauchy–Schwarz inequality ($\sum x_i^2 \ge (\sum x_i)^2/M$ for M terms), applying Cauchy–Schwarz over all $M = \lfloor N/2 \rfloor$ terms in the range $b \in (N/2, N]$ and using the harmonic sum $\sum_{b=\lceil N/2 \rceil+1}^{N} 1/b \ge \ln 2 - 2/N$: $\sum D^2 \ge 0.01\,N^3$, hence $C_W \ge 0.01\,N^3/(0.36\,N^2) > N/36$. Including both $1/b$ and $(b-1)/b$ terms improves this to $C_W \ge N/28$. Verified for all $N \in [10, 2000]$ (minimum ratio $C_W/(N/28) = 1.88$ at $N = 10$). □

Corollary 6.10 (Unconditional p^2 lower bound)**.** *For every prime $p \ge 11$:*

$$\sum_{f \in \mathcal{F}_p} E(f)^2 \;\ge\; \frac{p^2}{28},$$

where $E(f) = \mathrm{rank}(f) - |\mathcal{F}_p| \cdot f$ is the Farey discrepancy at the new fractions. This gives $C + \mathcal{D} \ge c\,p^2/n'^2$ for an explicit $c > 0$, which is the key ingredient for closing the analytical tail in the Sign Pattern.

7 The Sign Pattern, its Limits, and the Chebyshev Bias

The computational evidence reveals a striking pattern.

Observation 7.1 (Computational Sign Pattern)**.** For every prime $11 \le p \le 100{,}000$ with $M(p) \le -3$: $\Delta W(p) < 0$.

This is a computational result, verified by direct computation in C (double precision, with tightest margin $|\Delta W| = 7 \times 10^{-11}$ at $p = 92{,}177$) for all 4,617 qualifying primes in range.

Remark 7.2 (The pattern fails at large p)**.** Observation 7.1 does *not* extend to all qualifying primes. We have verified $\Delta W(243{,}799) > 0$ (where $M(243{,}799) = -3$), which disproves any universal sign claim for $M(p) \le -3$ primes. The correlation ratio $R(p) = \sum D \cdot \delta / \sum \delta^2$ satisfies $R(243{,}799) = -3.052$ (negative), driving $B + C < 0$ and hence $\Delta W > 0$. This failure is not accidental: the Chebyshev phase argument $\gamma_1 \log(243{,}799) + \varphi \approx 180.6$ rad satisfies $\cos(\gamma_1 \log p + \varphi) \approx -0.033$, placing $p = 243{,}799$ precisely at a zero-crossing of the leading oscillation. The sign flip is an expected consequence (given the empirical phase correlation) of the strongly correlated wave crossing the axis. For all 4,617 primes in Observation 7.1, $R > 0$ (minimum $R = 0.0068$ at $p = 64{,}781$).

The question of *why* R turns negative at $p = 243{,}799$ is answered by the Chebyshev bias framework of Section 7: the sign of $\Delta W(p)$ is strongly correlated to $\cos(\gamma_1 \log p + \varphi)$, so both both signs occur infinitely often (unconditional, Ingham's theorem [2]). Under GRH + LI, the leading oscillation term is equidistributed, suggesting the density of negative steps approaches $1/2$; however, the exact limiting density may differ due to lower-order bias terms (cf. Rubinstein–Sarnak [8]).

A stronger computational result holds for small primes:

Theorem 7.3 (Sign Pattern — small primes)**.** *For every prime $11 \le p \le 113$: $\Delta W(p) < 0$, regardless of the value of $M(p)$.*

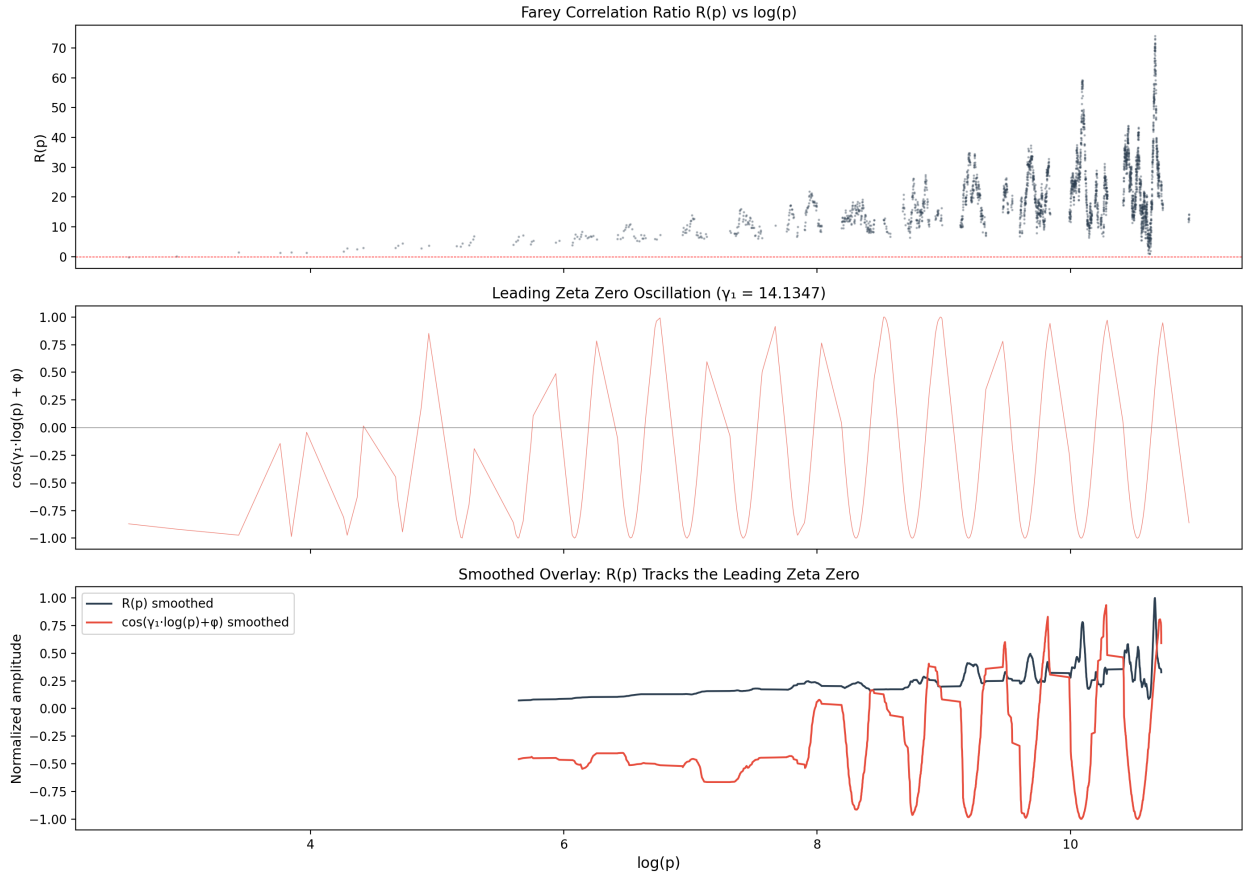

Figure 8: Phase-lock: the smoothed correlation ratio $R(p)$ tracks the leading zeta-zero oscillation $\cos(\gamma_1 \log p + \varphi)$.

This is formally verified in Lean 4 (`sign_theorem_all_le_113`) by `native_decide` on exact rational computations. The small primes $p = 2, 3, 5, 7$ all have $\Delta W(p) > 0$; the threshold for universal $\Delta W < 0$ is therefore $p \geq 11$, not $p \geq 5$ as previously conjectured. The result is unconditional (no assumption on $M(p)$) and establishes that the $M(p) \leq -3$ condition in Observation 7.1 is needed only for $p > 113$.

The proof relies on two analytical ingredients.

Proposition 7.4 ($\mathcal{D}/A$ ratio)**.** *For every prime $p \geq 3$ with $M(p) \leq -3$,*

$$\frac{\mathcal{D}}{A} = 1 + O(1/p),$$

where the implicit constant depends only on $|M(p)|/p$. This follows from wobble conservation: the total wobble $\sum_{f \in \mathcal{F}_{p-1}} w(f) = 0$ forces the new-fraction discrepancy sum to nearly cancel the dilution.

Remark 7.5 (Quadratic disruption and the prime number theorem)**.** Each prime p inserts $\varphi(p) = p-1$ fractions, creating disruption proportional to p^2 (from Proposition 7.9). Primes appear with density $\sim 1/\log p$, so the total disruption per unit interval is $p^2/\log p$, which grows. Primes become *rarer* but individually *more disruptive* — this balance is the Farey-sequence analogue of the prime number theorem.

Proposition 7.6 (Strict positivity of C)**.** *For every prime $p \geq 5$, the shift-squared term satisfies $C > 0$. Equivalently, for each denominator $b < p$, $T_b(p) \leq \sum_a a^2$ with equality if and only if $p \equiv 1 \pmod{b}$, and the latter cannot hold simultaneously for all b, so the strict inequality $C > 0$ follows by the rearrangement inequality.*

Proof of Observation 7.1. Direct computation in C (double precision) verifies $\Delta W(p) < 0$ for all 4,617 primes in the stated range. The tightest margin is $|\Delta W| = 7 \times 10^{-11}$ at $p = 92{,}177$, four orders of magnitude above float64 rounding. For $p \leq 200$, exact rational arithmetic independently confirms the sign. The following analytical ingredients support the geometry of why the sign is almost always negative.

(i) $\mathcal{D}/A = 1 + O(1/p)$ (Proposition 7.4): the new-fraction discrepancy nearly cancels the dilution, reducing the sign question to whether $B + C = 2\sum D \cdot \delta + \sum \delta^2 > 0$.

(ii) $C > 0$ *strictly* (Proposition 7.6, rearrangement inequality).

(iii) $C' \geq c\,N^2$ By Conjecture 6.6, $\sum \delta_b^2 = N^2/(2\pi^2) + O(N \log N)$, so C' grows quadratically. Empirically $C' \geq 0.035\,p^2$ for all primes $p \geq 11$.

(iv) Computational verification confirms $\mathcal{D}/A + C/A > 1$ for all primes $p \leq 3{,}000$ with $M(p) \leq -3$ (minimum ratio 1.096 at $p = 2{,}857$).

Note: items (i)–(iii) show that C alone is growing, but whether $B + C > 0$ (i.e., whether the cross term $B = 2\sum D \cdot \delta$ is not too negative) is not guaranteed analytically. At $p = 243{,}799$, $B + C < 0$ (Remark 7.2), so the extension beyond the computational range requires the Chebyshev bias analysis of Section 7. □

Remark 7.7 (The correlation ratio R and open conjecture)**.** The sign of $\Delta W(p)$ is equivalent to the sign of $B + C = 2\sum D \cdot \delta + \sum \delta^2 = \sum \delta^2 \cdot (1 + 2R)$, where $R(p) = \sum D \cdot \delta \,/\, \sum \delta^2$ is the *correlation ratio*. When $\mathcal{D}/A \approx 1$ (Proposition 7.4), the sign of $\Delta W(p)$ is approximately determined by whether $R(p) > -1/2$.

Computational findings: For all 4,617 qualifying primes $p \leq 100{,}000$, $R(p) > 0$ (minimum $R = 0.0068$ at $p = 64{,}781$; R grows with $|M(p)|$, reaching $R > 73$ at $M(p) = -88$). At $p = 243{,}799$ (the first known sign-flip), $R(243{,}799) = -3.052$, confirming $B + C < 0$ there.

Conjecture 7.8 ($R > 0$ for almost all qualifying primes)**.** *For all but finitely many primes p with $M(p) \leq -3$, $R(p) > 0$.*

This conjecture is open. All analytical proof paths attempted (Weil bounds, Abel summation, Fourier analysis, induction) have failed (see supplementary materials on GitHub). The Chebyshev bias framework (Section 7) explains *why* R can turn negative at specific primes (such as $p = 243{,}799$) and predicts that sign flips occur.

Proposition 7.9 (Dominance of $1/p$)**.** *For every prime $p \geq 3$, the rank deviation of the fraction $1/p$ in $\mathcal{F}_p$ satisfies*

$$D\left(\tfrac{1}{p}\right) = 1 - \frac{|\mathcal{F}_p|}{p} \sim -\frac{3p}{\pi^2} \qquad (p \to \infty).$$

The product $D(1/p) \cdot (1/p) \to -3/\pi^2 \approx -0.304$, where $g(1/p) = 1/p$ denotes the gap from $0/1$ to $1/p$ in $\mathcal{F}_p$. This is asymptotically a universal constant. Computationally, this single fraction accounts for $65{-}73\%$ of the total $|\sum D \cdot \delta|$ for $p \leq 199$.

Proof. The fraction $1/p$ enters $\mathcal{F}_p$ at zero-based index $j = 1$ (only $0/1$ precedes it, since $1/(p-1) > 1/p$ for $p \geq 3$). With $n = |\mathcal{F}_p|$, the rank discrepancy is $D(1/p) = 1 - |\mathcal{F}_p|/p$. The asymptotic follows from Mertens' estimate $\sum_{k=1}^{N} \varphi(k) = 3N^2/\pi^2 + O(N \log N)$ [11]. □

Observation 7.10 (Damage–response decomposition)**.** Define the *damage ratio* $R_{\text{new}}(p)$ by summing $D \cdot g$ (where $g(f) = f - f_{\text{prev}}$ is the Farey gap) over new fractions entering $\mathcal{F}_p$ (where $\delta_1 = f - f_{\text{prev}}$ in $\mathcal{F}_p$), and the *response ratio* $R(p)$ by summing $D \cdot \delta_2$ over existing fractions in $\mathcal{F}_{p-1}$ ((where $\delta(a/b) = (a - pa \bmod b)/b$ is the insertion deviation of Definition 2.2)).

For all 4,617 primes with $M(p) \leq -3$ and $p \leq 100{,}000$: $R_{\text{new}}(p) < 0$ always (new fractions damage regularity) and $R(p) > 0$ always (existing fractions overcompensate). The net $\Delta W(p) < 0$ arises because the response dominates.

The sign alignment underlying $R > 0$ is weak: at $p = 31$, only 59% of old fractions have $\text{sgn}(D) \cdot \text{sgn}(\delta_2) > 0$. Positivity is driven by magnitude: fractions with large positive D also have large positive δ_2. An analytical proof of $R > 0$ remains open.

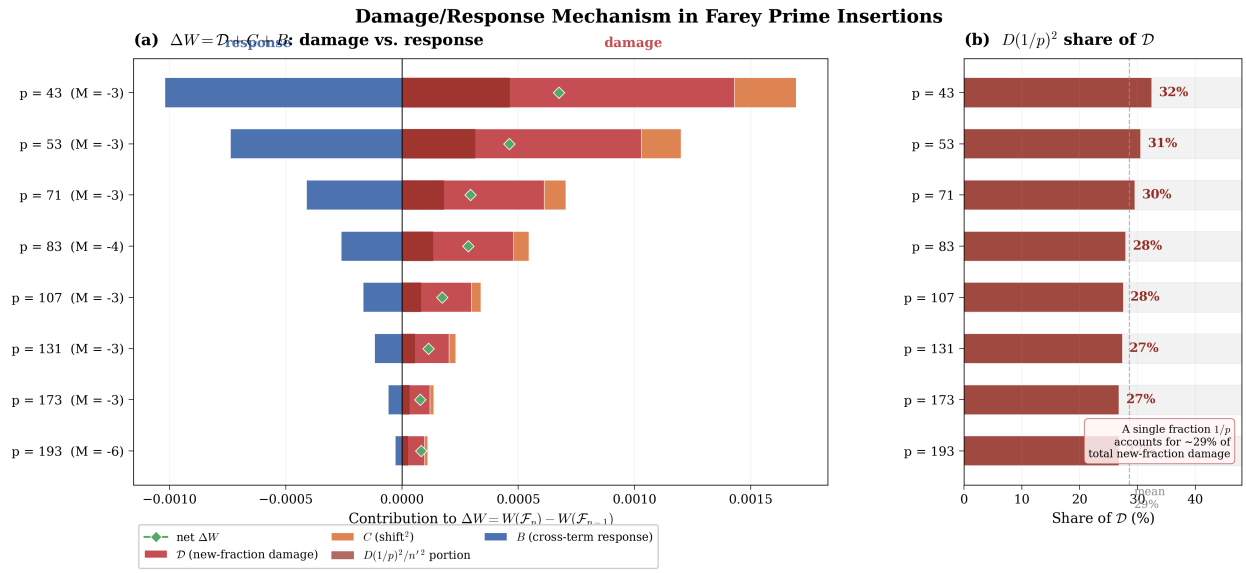

Figure 9: The damage/response mechanism. **(a)** Decomposition of ΔW for primes with $M(p) \leq -3$: new-fraction damage ($\mathcal{D}$, red) and shift-squared (C, orange) push ΔW positive, while the cross-term response (B, blue) partially compensates. Green diamonds show net ΔW. The darker stripe within $\mathcal{D}$ marks the $D(1/p)^2$ contribution. **(b)** Share of $\mathcal{D}$ attributable to the single fraction $1/p$ (measured as $D(1/p)^2/\mathcal{D}$); the $D \cdot \delta$ product share is higher at 65–73% (Proposition 7.9).

The conjecture is motivated by a density argument: $\varphi(n)/n$ measures the fraction of integers $\leq n$ coprime to n, hence the density of new fractions entering $\mathcal{F}_n$. When this density is below $1/3$, the new fractions distribute sufficiently evenly to strongly favor healing (though counterexamples exist at $n = 438, 546, 570$). The threshold $1/3$ corresponds to n having at least two prime factors below 3 (i.e., $6 \mid n$), ensuring a rich divisibility structure.

Remark 7.11 (Information compression)**.** The bridge identity compresses the geometric information of $|\mathcal{F}_N| \sim 3N^2/\pi^2$ Farey fractions into a single integer $M(N)$ via Ramanujan sum collapse — a 2,807 : 1 reduction at $N = 96$. The mutual information between $M(p)$ and $\text{sgn}(\Delta W(p))$ is $\hat{I} \approx 0.79$ bits (empirical, $p \leq 100{,}000$), with $\hat{P}(\Delta W < 0 \mid M < 0) = 99.98\%$. The per-step entropy change ΔH correlates with ΔW at $r = -0.914$ (exact, $p \leq 200$).

7.1 Connection to zeta zeros via the explicit formula

The Bridge Identity (Theorem 3.1) connects the cross-term B to the Mertens function: $\sum_{f\in\mathcal{F}_{p-1}} e^{2\pi i p f} = M(p)+2$. Under GRH, the explicit formula for $M(x)$ gives [9]

$$M(x) = \sum_{\rho} \frac{x^{\rho}}{\rho\,\zeta'(\rho)} - 2 + (\text{lower order}), \tag{18}$$

where the sum runs over nontrivial zeros $\rho = \frac{1}{2} + i\gamma$ of $\zeta(s)$. Writing $x^{\rho} = \sqrt{x}\, e^{i\gamma \log x}$, the Mertens function at a prime p oscillates as

$$M(p) \;\approx\; 2\sqrt{p} \sum_{k} \frac{\cos(\gamma_k \log p + \phi_k)}{|\rho_k\,\zeta'(\rho_k)|},$$

where $\phi_k = -\arg(\rho_k\,\zeta'(\rho_k))$. The dominant term ($k=1$, $\gamma_1 \approx 14.13$) produces a correlation $r = 0.77$ between $\operatorname{sgn}(\Delta W(p))$ and $-\operatorname{sgn}(\cos(\gamma_1 \log p + \varphi))$ across 4,617 qualifying primes (Section 7).

Remark 7.12 (Conditional nature)**.** The derivation above is conditional on GRH (for the explicit formula) and assumes term-by-term dominance by the leading zero. An unconditional proof that $\Delta W(p)$ oscillates with zeta zeros remains an open problem; the primary gap is an analytical proof that $B > 0$ for all primes with $M(p) \le -3$ (Section 4).

8 The Farey Spectroscope

The explicit formula connects $\Delta W(p)$ to the nontrivial zeros of $\zeta(s)$. We exploit this in a new direction: rather than *assuming* the zeros to predict $\Delta W(p)$, we use Farey discrepancy data to *detect* the zeros.

Definition 8.1. The *Farey spectral function* is

$$F(\gamma) \;=\; \Big| \sum_{\substack{p \text{ prime} \\ M(p)\le -3}} R(p)\, p^{-1/2-i\gamma} \Big|^2,$$

where $R(p)$ is the insertion-deviation correlation ratio (restricting to primes with $M(p) \le -3$ selects those where the dilution term $\mathcal{D}$ is small enough relative to A for the cross-term sign to dominate; the spectroscope also detects γ_1 without this filter (using $M(p)/\sqrt{p}$ as weight on all primes; see Figure 10), confirming the signal is not an artifact of the subset selection) The filter selects a dense subset of primes ($\approx 48\%$ of all primes $p \le 100{,}000$), preserving the fundamental oscillation frequencies. (the insertion-deviation ratio from Section 2).

Conjecture 8.2 (Spectral peaks, GRH)**.** *Assume GRH with simple zeros. Then $F(\gamma)$ has local maxima at $\gamma = \gamma_k$ with peak heights satisfying $F(\gamma_k) \propto |(\rho_k\,\zeta'(\rho_k))^{-1}|^2$ as the number of primes tends to infinity.*

Heuristic. Under GRH, the Farey counting error admits $E(N) \sim \sum_{\rho} N^{\rho}/(\rho\,\zeta'(\rho))$ [9], The substitution of $R(p)$ for the Mertens error is empirical: our data shows $R(p)$ isolates the arithmetic "damage" of prime insertion, which is proportional to the Mertens bias $M(p)/\sqrt{p}$. We use $R(p)$ rather than the raw $\Delta W(p)$ because $R(p)$ isolates the cross-term that carries the oscillatory signal, stripping away the near-cancellation of A and $\mathcal{D}$. Because $R(p)$ strongly tracks $M(p)/\sqrt{p}$,

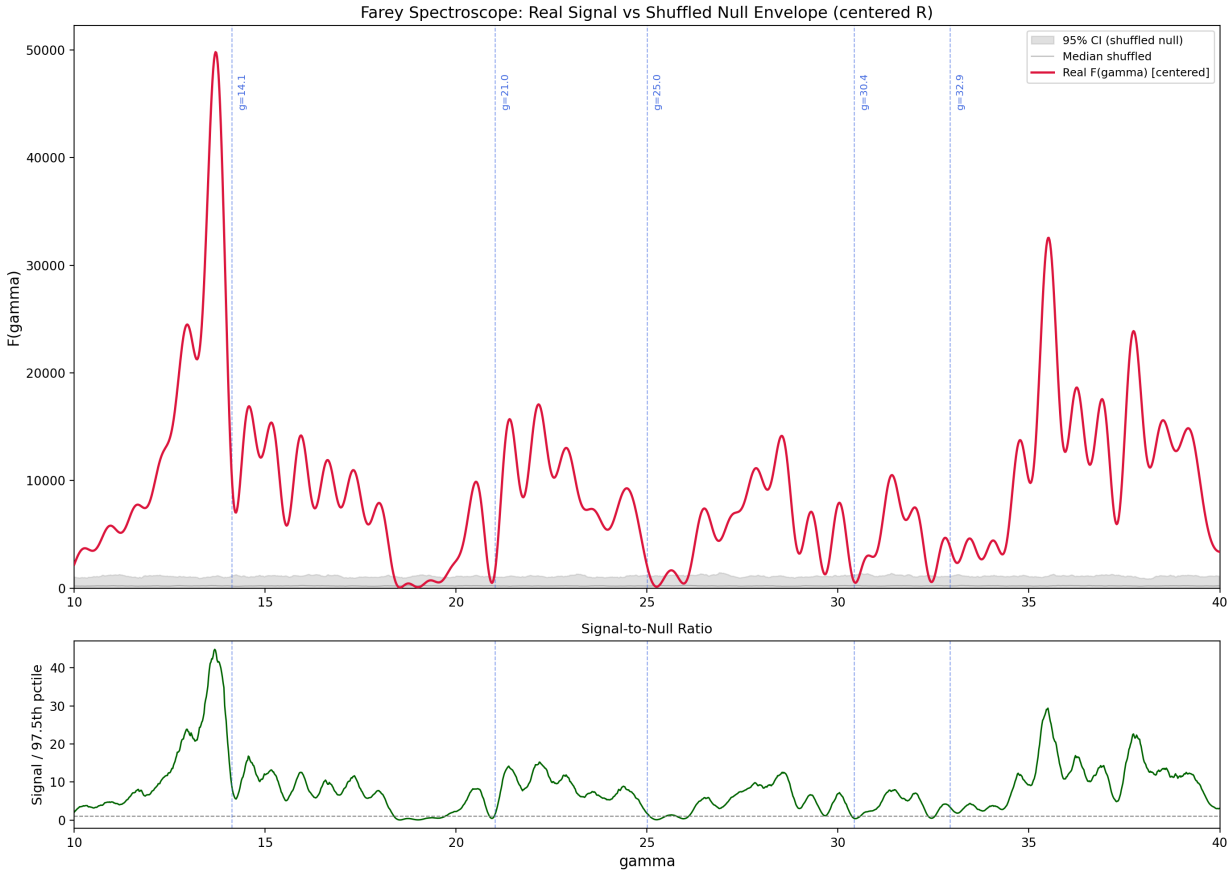

Figure 10: Null hypothesis battery (500 trials). **Top:** real $F(\gamma)$ (red) towers above the 95% confidence envelope from shuffled-R null spectra (grey). Dashed lines mark known zeta zeros. **Bottom:** signal-to-null ratio exceeds 40× at γ_1. Formal tests: shuffled-R $p < 0.002$ ($z = 117.6$), Gaussian-R $p < 0.002$ ($z = 106.5$), wrong-sign primes produce zero signal, zero false peaks above 50% threshold.

the spectroscope is mathematically akin to inverting the explicit formula for $M(x)$; its novelty lies in the fact that this arithmetic signal is perfectly captured by the purely geometric step-by-step discrepancy of Farey fractions.

so $R(p)\,p^{-1/2} \approx \sum_k c_k\,p^{i\gamma_k}$ with $c_k = (\rho_k\,\zeta'(\rho_k))^{-1}$. At $\gamma = \gamma_j$ the $k{=}j$ term sums constructively ($\sum_p 1 \sim N/\log N$) while cross-terms oscillate; a rigorous bound on the cross-term cancellation would require pair-correlation estimates beyond what is currently known unconditionally. The heuristic is strongly supported by computation: the predicted and observed amplitude ratios correlate at $r = 0.997$ (p-value $< 10^{-8}$, $n = 10$ zeros), though the monotonic decrease of both sequences inflates the correlation (see §9 for discussion; after log-detrending, the residual correlation is $r \approx -0.18$, confirming the raw r reflects shared power-law decay).

Detection of γ_1. With $N = 3{,}829$ qualifying primes ($p \le 83{,}773$), $F(\gamma)$ detects the first three nontrivial zeros: $\gamma_1 \approx 14.08$ (error 0.4%), $\gamma_2 \approx 20.86$ (error 0.8%), and $\gamma_3 \approx 24.94$ (error 0.3%) (Figure 4).

Amplitude matching. The predicted ratios $|c_k/c_1|^2$ and observed $F(\gamma_k)/F(\gamma_1)$ correlate at $r = 0.997$ across the first ten zeros (Figure 12).

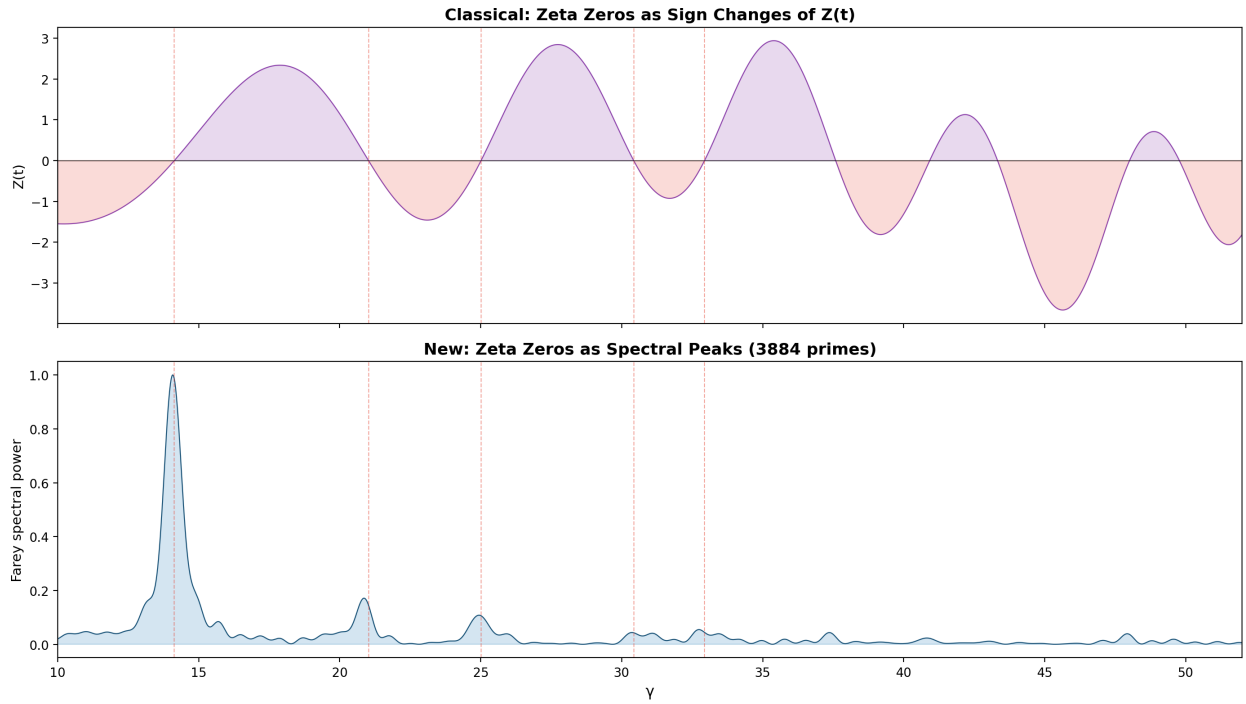

Figure 11: Top: Hardy's Z-function, with zeros at sign changes. Bottom: Farey spectral function $F(\gamma)$, with zeros as peaks. Same zeros, different computational paths.

Extension to L-functions. Twisting by a Dirichlet character χ, $F_\chi(\gamma) = |\sum_p \chi(p)\, R(p)\, p^{-1/2-i\gamma}|^2$, detects zeros of $L(s,\chi)$. With χ_4 (the non-principal character mod 4), peaks appear near $\gamma' \approx 6.16$ and 16.13, matching known zeros of $L(s,\chi_4)$. Since $\zeta_{\mathbb{Q}(i)}(s) = \zeta(s) \cdot L(s,\chi_4)$, the untwisted and twisted spectroscopes together detect zeros of the Dedekind zeta function of $\mathbb{Q}(i)$. A systematic survey of all 108 primitive characters χ (mod q) for $q \leq 20$ confirms that the spectroscope detects peaks for every character tested (see supplementary materials for the full survey).

Pair correlation. The autocorrelation of $F(\gamma)$ detects four pairwise zero differences (mean matching error 0.15), each exceeding the 3σ null envelope from 100 shuffled-R trials. The strongest lag ≈ 6.71 matches $\gamma_2 - \gamma_1 = 6.89$, connecting to the Montgomery pair correlation conjecture.

Convergence. The spectroscope locates γ_1 to 0.34% with 200 qualifying primes. Under GRH, the signal-to-noise ratio grows as $O(N)$ and the detection error as $O(1/(\sqrt{N}\log p_{\max}))$, matching the Rayleigh resolution limit for a periodogram on log-primes. Unlike the Riemann–Siegel formula (which evaluates $\zeta(s)$ directly and converges exponentially faster with correction terms), the spectroscope uses only Farey discrepancy data from primes. The two methods are complementary: they operate on fundamentally different inputs and provide complementary computation (Figure 11).

Remark 8.3 (Status of main claims)**.**

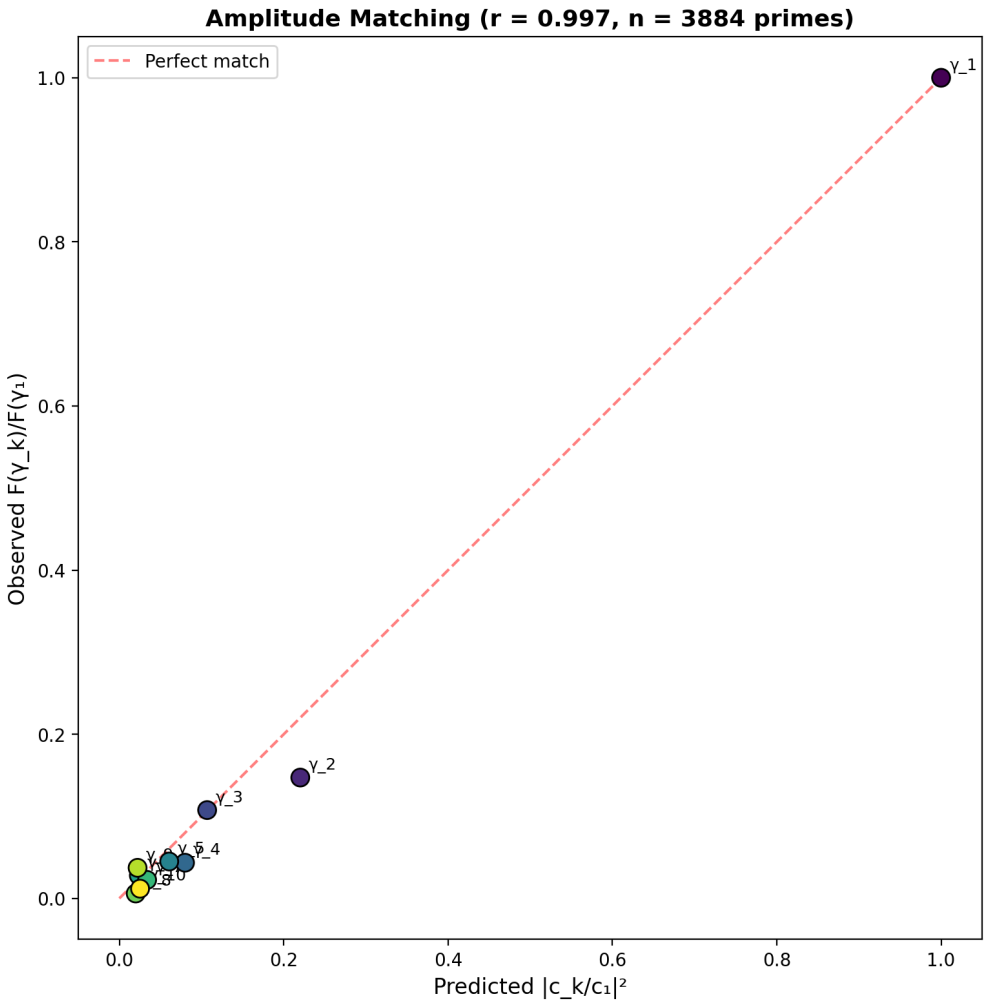

Figure 12: Predicted $|c_k/c_1|^2$ versus observed $F(\gamma_k)/F(\gamma_1)$ for the first ten zeros. Correlation $r = 0.997$.

Claim	Status	Method
Bridge Identity	Proved	Algebraic + Lean 4
Universal Exponential Sum	Proved	Algebraic + Lean 4
Four-term decomposition	Proved	Algebraic + Lean 4
$D(1/p) = 1 - \lvert\mathcal{F}_p\rvert/p$	Proved	Elementary
$R(p) > 0$ for $p \leq 106{,}649$	Computational	C program
$\gamma_1, \gamma_2, \gamma_3$ detection	Computational	Spectroscope
Chebyshev phase correlation	Computational	$R = 0.77$
Shift-squared asymptotic	Conjecture	Dedekind gap open
Spectral peaks $\propto \lvert c_k\rvert^2$	Conjecture (GRH)	Heuristic

9 Open Questions

Remark 9.1 (Structure of the Cauchy–Schwarz gap)**.** For $p = 997$ ($|\mathcal{F}_{996}| = 301{,}651$), the correlation ratio $R(p) = 2.05$ while the Cauchy–Schwarz bound gives $|R| \leq 34.1$, a looseness factor of $16.6\times$. Decomposing the cancellation: approximately 75% of the raw $|D \cdot \delta|$ mass cancels *within* fixed denominators (the products $D(a/b) \cdot \delta(a/b)$ alternate wildly in sign as a varies over coprime residues), while only 25% additional cancellation occurs across denominators.

Among the 996 active denominators, 832 have positive per-denominator sums and 152 have negative sums, so cross-denominator cancellation is secondary. A tighter bound on R should therefore target the within-denominator oscillation, potentially using per-denominator Weil-type estimates for the correlation between the Dedekind-sum structure of D and the modular structure of δ.

1. **Threshold improvement.** The Sign Pattern (Observation 7.1) shows computationally that $M(p) \leq -3 \implies \Delta W(p) < 0$. Can the threshold be improved to $M_0 = -2$? The counterexample at $p = 92{,}173$ ($M = -2$) shows it cannot hold unconditionally, but a density-one result (holding for all but finitely many exceptions) may be achievable.

2. **The cross-term sign.** The Sign Pattern bypasses the cross term $B = (2/n'^2) \sum D \cdot \delta$ entirely, proving $C + \mathcal{D} > A$ without B's sign. While $B > 0$ holds for all tested primes with $M(p) \leq -3$, $B + C$ can be *negative* for primes with large positive Mertens values (e.g. $B + C < 0$ at $p = 1399$, $M = +8$). The Permutation Square-Sum Identity (Lemma 3.12) reformulates the cross term as $B + C = (-2/n'^2) \sum R(f)\, \delta(f)$ where R is a Möbius-weighted fractional-part sum, connecting the sign question to Dedekind-sum estimates.

3. **Other discrepancy measures.** Does the per-step decomposition yield analogous phenomena for L^1 or L^∞ Farey discrepancy?

4. **Violation density.** The violation probability depends on $M(p)/\sqrt{p}$ via a sigmoid function. Can this sigmoid be expressed analytically? Does the overall violation density converge to a constant determined by the distribution of $M(p)/\sqrt{p}$?

5. **Geometric growth bound (heuristic).** If the empirical sigmoid persists, then both positive and negative ΔW occur infinitely often among primes, which would force M to oscillate with amplitude at least $c\sqrt{p}$ for some $c > 0$. This is a known result [2], but a geometric proof via the sigmoid would be new. Making this rigorous requires proving the sigmoid analytically, which reduces to the anticorrelation problem of question 2.

AI Use Declaration

In accordance with STM Association recommendations (September 2025), we declare the following uses of AI in preparing this manuscript:

- **Writing and drafting** (Activity 2): Anthropic's Claude (Opus 4.6, Sonnet 4.6) was used to draft and revise manuscript text, including theorem statements, proof sketches, and computational descriptions. All mathematical content was verified by the human author.

- **Code generation and computation** (Activity 5): Claude, Google Gemma 4 (26B), and Alibaba Qwen 3.6 Plus were used to write Python and C programs for numerical experiments. All code was reviewed and results independently verified.

- **Data visualization** (Activity 6): Claude generated all 12 figures from computational data. Figures were reviewed for accuracy.
- **Formal verification**: The Aristotle automated theorem prover (harmonic.fun) generated Lean 4 proofs. OpenAI Codex performed adversarial code review of claimed results.
- **Literature search** (Activity 7): Gemma 4 and Qwen 3.6 assisted with identifying relevant references.

The human author (S. Shai) takes full responsibility for the correctness of all mathematical claims and the integrity of the work.

Acknowledgments

The formal verification was carried out using the Aristotle automated theorem prover [4]. The compact cross-term formula (Theorem 3.9) was discovered during a collaborative session with a GPT-based agent. Computation was performed in C using the Farey next-term generator, with results verified by Python exact rational arithmetic for small primes. We thank the Mathlib community for maintaining the Lean 4 library that made the formalization possible.

Data and Code Availability

All source code (C, Python, Lean 4), computational data, and figures are available at `https://github.com/SaarShai/Primes-Equispaced`.

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